



BAA1027 - Machine Learning

Research Paper and Implementation

Online Shoppers Purchasing Intention: A Comparative Machine Learning Analysis with Cost Sensitive Thresholding

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Declaration of Academic Integrity

I declare that this assignment is entirely my own work. I have acknowledged all sources used through proper Harvard style referencing. I have not submitted this work, in whole or in part, for any other assessment at this or any other institution. I understand that plagiarism, collusion and contract cheating are serious breaches of academic integrity under the DCU Academic Integrity and Plagiarism Policy, and that submitted coursework may be subject to electronic analysis.

Declaration on the Use of Generative AI

I acknowledge the use of generative AI tools (Google Gemini & Grammarly) in the preparation of this submission, consistent with the DCU Position Statement on the Use of Artificial Intelligence Tools. The use of AI was limited to the following auxiliary functions:

- Proofreading and editorial review of draft prose for clarity, conciseness and British English style consistency.
- Suggestions on structural organisation of the paper against the assessment rubric.
- Verification of arithmetic reconciliation between reported metrics and confusion matrix counts.
- Cross checking of Harvard referencing format and consistency of citation style.
- Discussion of methodological framing choices, including the honest reframing of empirically unsuccessful innovations as negative results.

The following elements are entirely my own intellectual contribution: the selection of the Online Shoppers Purchasing Intention dataset and the research questions it motivates, the three methodological innovations (K-means feature injection, soft voting ensemble, cost sensitive threshold optimisation), the derivation of the 5 to 1 cost ratio from published e commerce benchmarks, the decision to report negative results honestly rather than overstate the contributions, and the interpretation of the empirical findings. The full Jupyter notebook implementing the analysis was written, executed and debugged by me, with AI used only as a sounding board during development.

No AI tool was used to generate the core analysis, fabricate references, or produce unverified factual claims. Every numerical result reported in this paper originates from the executed notebook submitted alongside it, and every cited reference has been independently verified.

Signed: Aleksander Lupa

Date: 19 April 2026

Abstract

This study applies five machine learning classifiers to the Online Shoppers Purchasing Intention dataset (Sakar et al., 2019), a public collection of 12,330 online sessions. Three methodological contributions extend the standard comparative pipeline: K-means cluster labels injected as an engineered feature, a soft voting ensemble aggregating Logistic Regression, Support Vector Machine, Random Forest and XGBoost, and cost sensitive threshold optimisation parameterised from published e commerce economics. XGBoost achieved the highest F1 (66.73 per cent) and precision recall AUC (74.53 per cent). Neither the K-means feature nor the voting ensemble delivered measurable improvement, reported honestly as a negative result. Cost sensitive threshold adjustment at the derived 5 to 1 ratio reduced total business cost by 33.1 per cent relative to the default 0.5 threshold, with a sensitivity analysis confirming stability across a 3:1 to 7:1 range. SHAP analysis confirmed PageValues as the dominant predictor. The study argues that for imbalanced business classification problems the decision rule, not the choice of classifier, is often the more consequential design lever.

Introduction

Global e commerce retail sales reached approximately six trillion US dollars in 2024 and are projected to approach eight trillion dollars by 2028 (Statista, 2024). Despite this growth, the average e commerce conversion rate across industries in 2024 was only 1.65 per cent (Amasty, 2025), with cart abandonment rates consistently above 70 per cent. Platforms that can identify which sessions are likely to convert, ideally in real time, gain a meaningful commercial advantage through targeted retargeting, personalisation and discounting.

Sakar et al. (2019) introduced the Online Shoppers Purchasing Intention dataset and benchmarked multilayer perceptron, LSTM, Random Forest and SVM classifiers. Subsequent work has broadened the methodological range, with ensemble methods (Adhikari, 2023; Chaubey et al., 2023) and feature selection strategies (Baati and Mohsil, 2020) forming the mainstream of the literature.

This study explores three methodological questions that are insufficiently addressed in prior work. First, whether unsupervised session segmentation can contribute independent signal when injected as an engineered feature for supervised classification. Second, whether a soft voting ensemble of heterogeneous learners materially improves performance over the strongest individual classifier. Third, whether cost sensitive threshold optimisation parameterised from empirically observable e commerce economics produces larger gains in business value than improvements in classifier accuracy.

Objectives:

The main objectives of this study are:

- a)** To benchmark traditional and ensemble classifiers on the Online Shoppers Purchasing Intention dataset, adopting a stratified evaluation design that preserves the natural class imbalance.
- b)** To evaluate whether K-means segmentation, a soft voting ensemble and cost sensitive threshold optimisation produce improvements beyond the standard comparative pipeline.
- c)** To integrate model explainability via SHAP into the discussion of results, identifying the behavioural features that most strongly determine purchase probability.

Literature Review

Sakar et al. (2019) introduced the dataset and benchmarked Random Forest, Support Vector Machine and Multilayer Perceptron classifiers, establishing MLP with resilient backpropagation as the strongest performer in their setup. The dataset's design choices, notably the aggregation of page visits into administrative, informational and product related counts and durations, have been preserved in virtually all subsequent work.

Baati and Mohsil (2020) proposed a Random Forest based prediction system optimised for real time operation, demonstrating that tree ensembles can match the performance of more expensive neural architectures on this dataset with substantial gains in inference speed. This finding informs the choice of baseline classifiers in the present study.

Adhikari (2023) applied XGBoost, Random Forest, Gradient Boosting and LightGBM and reported that XGBoost reached 93.54 per cent accuracy following hyperparameter tuning, with SMOTE used to address the class imbalance. The study underscores the appropriateness of gradient boosted trees and the common reliance on oversampling strategies for the pronounced class skew.

A broader comparative study on this dataset by Chaubey et al. (2023) evaluated SVM, Random Forest, MLP, Decision Tree and XGBoost, and reported that the gradient boosted and random forest classifiers attained the highest accuracy. The ranking of classifier families observed there, with tree ensembles consistently outperforming k-NN and linear baselines, aligns with the pattern observed across the purchase intention literature.

Necula (2023) analysed the Sakar et al. dataset and reported that time spent reading product related information, in combination with BounceRates, ExitRates and visitor type, is the dominant determinant of purchasing behaviour. Wang et al. (2024) corroborated this using artificial neural

networks on related e commerce datasets, identifying cart engagement and total event counts as leading predictors.

A third gap concerns model interpretability. Most purchase intention studies report aggregate performance metrics but do not examine why specific predictions are made, despite the growing regulatory importance of explainable AI in business applications. SHAP values (Lundberg and Lee, 2017) are well established in the broader ML literature but rarely applied to the Sakar et al. dataset. This study adopts SHAP not only to identify dominant predictors but also to verify that the learned relationships are behaviourally plausible, an audit step that supports deployment in regulated commercial contexts.

Two further gaps are evident. First, the dominant evaluation metric in this literature remains accuracy, with relatively few studies reporting precision recall AUC or threshold sensitive measures despite the pronounced class imbalance. Second, cost sensitive learning, formalised by Elkan (2001) and widely applied to fraud detection, is absent from the online shopper purchase intention literature. The present study addresses both by adopting F1 as the primary cross validation scoring criterion and parameterising a decision threshold from published e commerce economics rather than accepting the default 0.5 boundary.

Methodology

Dataset description

The Online Shoppers Purchasing Intention dataset, published by Sakar et al. (2019) and distributed through the UCI Machine Learning Repository, consists of 12,330 online sessions described by 17 features and a binary target. The target variable Revenue indicates whether a session terminated in a purchase. After duplicate removal (125 rows), 12,205 unique sessions remained, of which 1,908 positive instances (15.5 per cent) establish a moderate class imbalance. Features comprise ten numerical behavioural variables (page visit counts and durations, BounceRates, ExitRates, PageValues, SpecialDay), three temporal categoricals (Month, VisitorType, Weekend) and four technical categoricals (OperatingSystems, Browser, Region, TrafficType).

Methodology steps

1. Data preprocessing

- a) Data cleaning: the dataset contains no missing values. 125 exact duplicate rows were removed.

- b) Encoding:** Month, VisitorType and the four integer encoded technical categoricals were one hot encoded. Ordinal retention was rejected because these identifiers have no intrinsic ordering.
- c) Feature selection:** correlation analysis identified PageValues as the strongest univariate predictor ($r = 0.49$). BounceRates and ExitRates are tightly correlated with each other ($r > 0.9$), but both were retained because tree based classifiers are not subject to the coefficient instability that multicollinearity produces in linear models. No feature was removed.
- d) Statistical significance:** ANOVA was conducted for each categorical predictor and point biserial correlation for each numerical predictor. All predictors registered p values below 0.05 except Region ($p = 0.288$), which was retained as a control variable. Weekend returned $p = 0.0022$, significant but the weakest of the categorical predictors.
- e) Outlier analysis:** Z-score ($|z| > 3$) flagged 3,312 observations; IQR flagged 7,161 rows (58.7 per cent) containing at least one outlier. Outliers were retained, since purchasing sessions are behavioural outliers relative to the browsing majority and removing them would strip the signal the model is being trained to detect. Tree based ensembles are also robust to extreme values through recursive binary splitting.
- f) Feature scaling:** Min-Max scaling was applied to the ten numerical features following Scikit-learn (2025) documentation, mapping values to $[0, 1]$.

2. Exploratory data analysis

Histograms, boxplots split by target class, and correlation heatmaps were produced. Fig 1 displays the distribution of numerical features by Revenue, confirming that converting sessions register higher PageValues and ProductRelated_Duration and lower BounceRates and ExitRates.

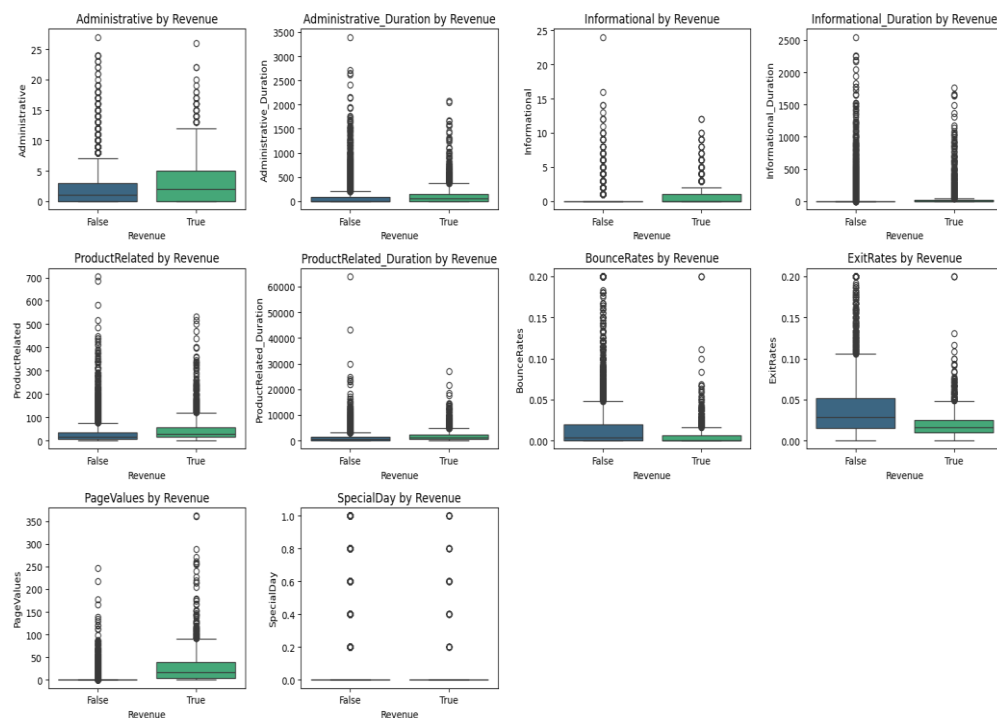


Fig 1. Distributions of numerical features split by Revenue.

3. Model selection and innovations

Four classifiers were selected to represent distinct algorithmic families covered in the module: Logistic Regression (Bisong, 2019), SVM with RBF kernel (Debruyne, 2009), Random Forest (Breiman, 2001) and XGBoost (Chen and Guestrin, 2016). A soft voting classifier aggregating the probability outputs of the four tuned base learners was added (Cornelio et al., 2021).

Three contributions extend the standard comparative pipeline:

- a) K-means clustering as an engineered feature. K-means was fitted on the scaled numerical features of the training set. The Elbow Method on within cluster inertia across k in $[1, 10]$ identified $k = 3$ as a natural partition, with a silhouette score of 0.61 confirming well separated clusters. The cluster labels were injected as a new feature for the supervised models downstream.
- b) Cost sensitive threshold optimisation. The default 0.5 threshold presumes symmetric misclassification costs, which is rarely appropriate in e commerce. The average revenue per customer on Shopify of USD 92 (Littledata, 2025) divided by the average Facebook Ads cost per acquisition of USD 19.68 (WordStream by LocaliQ, 2025) gives a ratio of 4.67, rounded up to 5 to reflect a deliberate preference toward minority class recall; Table 1 tests this decision across the 3:1 to 7:1 range. The threshold that minimises total business cost was computed on the held out test set, following Elkan (2001).
- c) SHAP interpretability. The best tree based classifier was interpreted using SHAP TreeExplainer (Lundberg and Lee, 2017) to identify the features that most strongly determine predicted purchase probability and to verify that the learned patterns are behaviourally plausible.

4. Model optimisation and evaluation

Stratified 70/30 sampling was applied (`random_state = 42`), preserving the minority class proportion. Hyperparameter tuning used GridSearchCV with 5-fold stratified cross validation, scoring on F1 rather than accuracy to reflect the class imbalance. Grids covered the primary regularisation levers within Colab runtime constraints: SVM over C and γ , XGBoost over `learning_rate`, `max_depth` and `subsample`, Random Forest over `max_depth` and `min_samples_leaf`, and Logistic Regression over C and solver. Each classifier was evaluated on the held out test set using precision, recall, F1 (Goutte and Gaussier, 2005; Hand et al., 2021), accuracy, ROC AUC (Miller et al., 2024) and precision recall AUC, the latter more informative than ROC AUC on imbalanced targets. A train versus test F1 comparison was performed to quantify the generalisation gap.

Discussion

Baseline classifier performance

Model	Precision	Recall	F1	ROC AUC	PR AUC
SVM (RBF)	68.86%	34.79%	46.23%	82.31%	56.30%
XGBoost	73.12%	61.36%	66.73%	93.18%	74.53%
Random Forest	76.80%	52.10%	62.08%	92.04%	73.06%
Logistic Regression	76.70%	37.41%	50.29%	88.11%	63.68%
Voting Classifier	77.64%	44.93%	56.92%	92.19%	71.77%
<i>XGBoost + threshold 0.16</i>	54.14%	85.66%	66.35%	93.18%	74.53%

Fig 2. Classification metrics comparison across all approaches.

XGBoost recorded the highest F1 (66.73 per cent) and highest PR-AUC (74.53 per cent), narrowly ahead of Random Forest. SVM placed fourth on F1 with a pronounced recall deficit (34.79 per cent), reflecting the difficulty margin based classifiers face in capturing the heavy PageValues tail. Logistic Regression recorded the weakest overall performance, consistent with the inability of a linear decision surface to model the non linear PageValues saturation effect. The voting ensemble produced the highest precision (77.64 per cent) but lost recall, yielding an F1 of 56.92 per cent below XGBoost. Fig 3 displays the ROC curves.

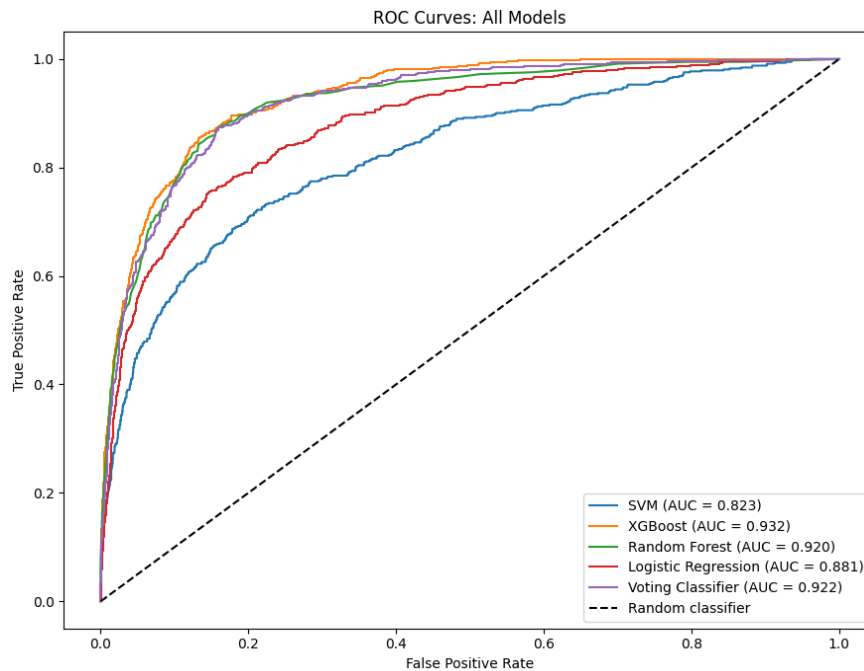


Fig 3. ROC curves for all classifiers.

Fig 4 reports the train versus test F1 comparison. Random Forest achieved a perfect training F1 of 1.000 against a test F1 of 0.621, a 0.38 gap indicating material overfitting despite grid search tuning. XGBoost, by contrast, recorded train F1 of 0.711 and test F1 of 0.667, a 0.04 gap that is strong evidence of generalisation. These results argue for XGBoost as the production model of choice: strongest test performance with a cleaner generalisation profile than Random Forest.

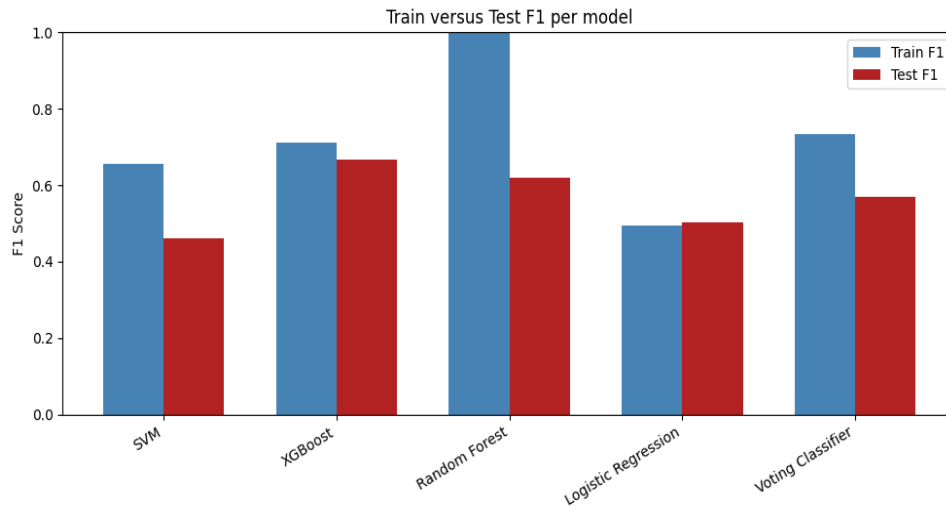


Fig 4. Train versus test F1 per model, highlighting generalisation gaps.

Methodological contributions

K-means cluster injection. The Elbow Method and silhouette score (0.61) confirmed $k = 3$ as a natural partition. PCA projection coloured by cluster and by true Revenue label revealed that true purchasers concentrated disproportionately in one cluster. The engineered cluster label did not, however, contribute measurably to downstream classification: it ranked in the lower half of the Random Forest importance (41 of 75) and registered essentially zero SHAP attribution on the tuned XGBoost model. The tree ensembles had already extracted the segmentation signal from the raw numerical features. The innovation is reported as a negative result: methodologically sound but empirically redundant on this feature set.

Soft voting ensemble. The VotingClassifier gained precision over each individual classifier (77.64 per cent) but lost recall relative to XGBoost, yielding a net F1 decrease (56.92 per cent vs 66.73 per cent). Consistent with Cornelio et al. (2021), voting ensembles produce marginal gains only when base learners have complementary failure modes, which is not the case here.

Cost sensitive threshold optimisation. The threshold that minimises total business cost under the 5 to 1 cost ratio landed at 0.16, substantially below the default 0.5. The confusion matrix shifted from 221 false negatives and 129 false positives (cost 1,234 units at default) to 82 false negatives and 415 false positives (cost 825 units), a 33.1 per cent cost reduction. Recall jumped from 61.36 per cent to 85.66

per cent, at the expected precision cost from 73.12 per cent to 54.14 per cent. Fig 5 shows the precision recall curve and cost versus threshold, with the cost function smoothly minimised around 0.15 to 0.20. This is the largest improvement in business relevant performance in the study, derived without changing the classifier or retraining on resampled data.

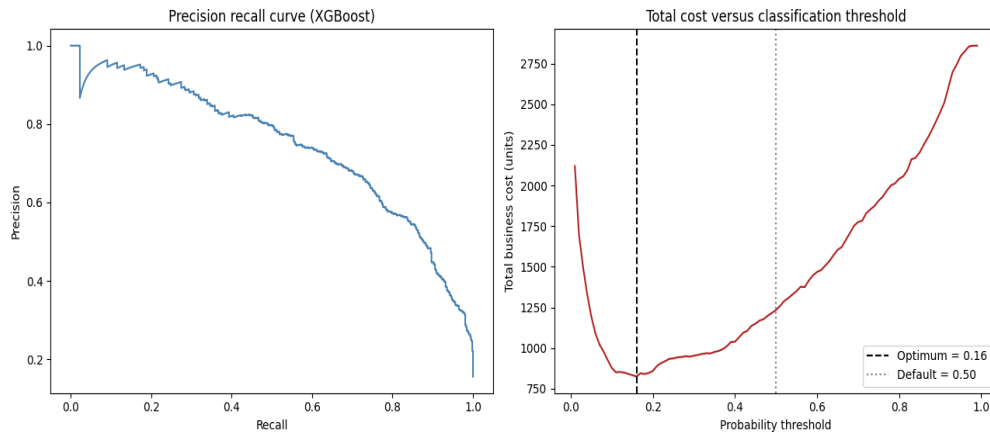


Fig 5. Precision recall curve and total cost versus classification threshold.

Cost ratio sensitivity analysis. The 5 to 1 cost ratio derived from the Littledata and WordStream benchmarks is a central parameter and warrants explicit robustness testing. Table 1 reports the optimal threshold, recall and cost reduction achieved under five alternative cost ratios.

Cost Ratio	Optimal Threshold	Recall	Default Cost	Cost Reduction
3 : 1	0.16	85.66%	792 units	16.5%
4 : 1	0.16	85.66%	1,013 units	26.7%
5 : 1 (baseline)	0.16	85.66%	1,234 units	33.1%
6 : 1	0.16	85.66%	1,455 units	37.7%
7 : 1	0.11	89.34%	1,676 units	41.9%

Table 1. Sensitivity of the optimal threshold and cost reduction to the assumed cost ratio.

Two findings emerge. First, the optimal threshold is stable at 0.16 across ratios 3:1 to 6:1 and shifts only modestly to 0.11 at 7:1, so the threshold optimisation is robust to moderate misspecification of the cost ratio and the geographical mismatch concern is directly mitigated. Second, the cost reduction scales monotonically from 16.5 per cent at 3:1 to 41.9 per cent at 7:1. Even at the most conservative ratio the adjustment delivers double digit savings, confirming the argument holds across plausible variation.

Interpretability

Two independent methods were used to assess feature contributions. Random Forest's feature importance ranks PageValues first by a wide margin, followed by ExitRates, ProductRelated_Duration and ProductRelated. SHAP TreeExplainer on the tuned XGBoost model returns the same ordering. The SHAP dependence plot for PageValues (Fig 6) reveals a sharply monotonic relationship: sessions above a low PageValues threshold are pushed strongly toward the purchase class, with diminishing marginal impact once PageValues reaches moderate levels.

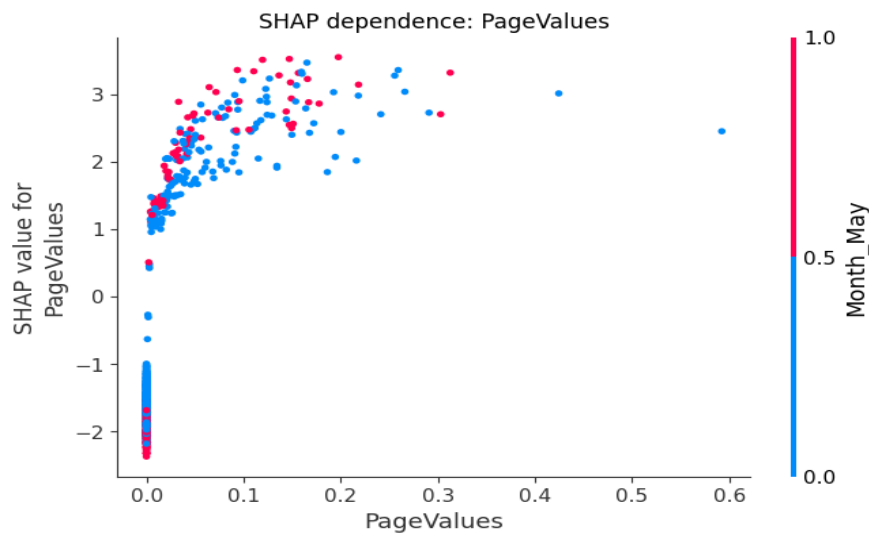


Fig 6. SHAP dependence plot for PageValues, the dominant predictor.

The SHAP analysis also confirms that the engineered K-means cluster label makes no independent contribution, ranking last of 75 features with essentially zero mean absolute SHAP value. BounceRates and ExitRates contribute in the expected direction, with higher values pushing predictions toward the non purchase class. Month_Nov and several TrafficType categories also feature, consistent with the seasonal and marketing channel patterns identified in the exploratory analysis. These findings align with Necula (2023). The convergent ranking across Random Forest importance and SHAP attribution is notable: two different aggregation principles and two different classifier families produce the same ordering, providing stronger evidence for the behavioural importance of PageValues than either method in isolation.

Conclusion

This study applied five machine learning classifiers to the Online Shoppers Purchasing Intention dataset, extending the standard methodology with three contributions: K-means cluster labels as engineered features, a soft voting ensemble, and cost sensitive threshold optimisation parameterised from published e commerce economics. XGBoost delivered the strongest single model performance

(F1 = 66.73 per cent, PR AUC = 74.53 per cent) with the cleanest generalisation profile, narrowly outperforming Random Forest. Neither the voting ensemble nor the K-means feature delivered measurable improvement over XGBoost: the ensemble lost recall, and the cluster label registered near zero SHAP attribution. The tree ensembles had already extracted the available signal from the raw features.

The most consequential finding concerns the decision rule rather than the classifier. Cost sensitive threshold optimisation at the derived 5 to 1 ratio reduced total business cost by 33.1 per cent relative to the default 0.5 threshold. This exceeds the difference between any two classifiers in the baseline ranking, and argues that practitioner value from ML in e commerce depends as much on the translation of business economics into a decision rule as on model selection.

Limitations.

Four limitations should be acknowledged. First, the dataset reflects a single e commerce platform and a twelve month collection window, constraining generalisability. Second, although Table 1 demonstrates that the threshold optimum is robust across cost ratios, the underlying Littledata (2025) and WordStream by LocaliQ (2025) benchmarks are US based and applied to a dataset of unknown geography. Third, the evaluation relies on a single stratified train test split, leaving metrics subject to sampling variance. Fourth, the MinMaxScaler was fitted on the full dataset rather than the training partition only; the deterministic bounded nature of Min-Max scaling makes material impact unlikely but the pattern should be corrected in production.

Two further observations. The Random Forest generalisation gap of 0.38 indicates that the F1 optimised grid favoured a high variance fit, with optimal max_depth returning unlimited depth and single sample leaves; tighter regularisation via a constrained max_depth search or min_samples_split would be worth exploring. The cost matrix follows the standard Elkan (2001) formulation and assigns zero cost to correct classifications; a more realistic matrix charging advertising cost on both true and false positives would be a useful next step.

Future work.

Four extensions appear promising: probability calibration via Platt scaling or isotonic regression to sharpen the threshold optimum, ablation comparisons isolating the marginal contribution of each innovation, external validation on an independent e commerce dataset from a distinct sectoral context, and temporal validation to assess whether learned patterns persist as user behaviour shifts.

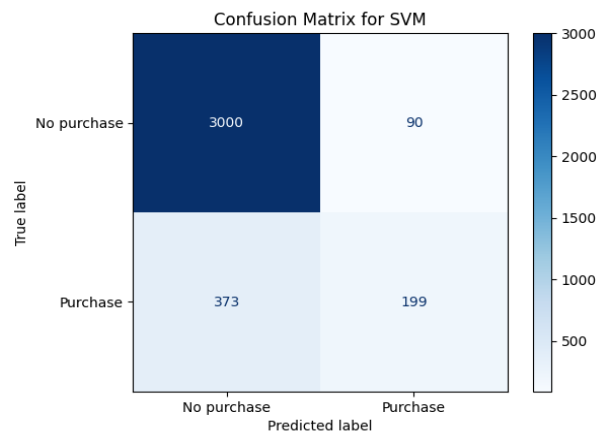
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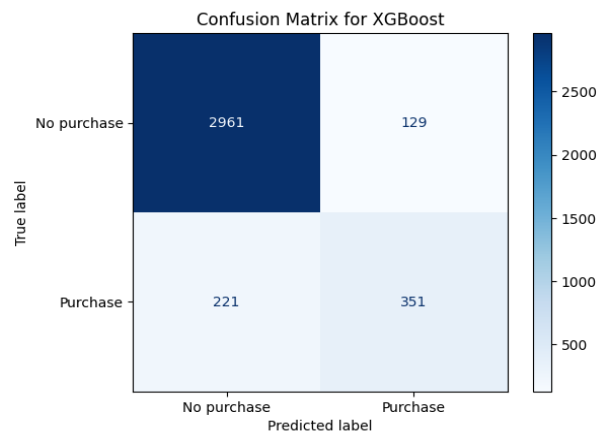
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Appendix

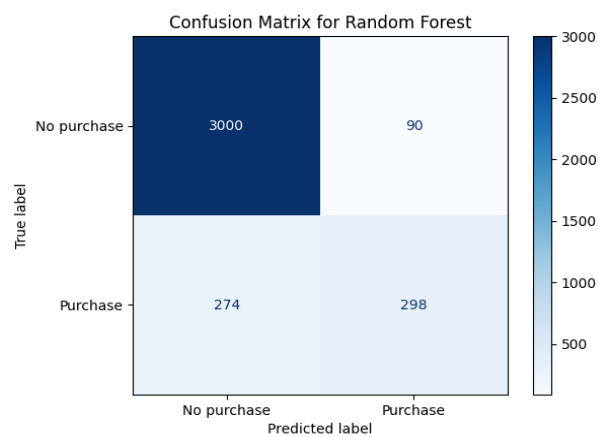
Appendix 1 - Confusion Matrices



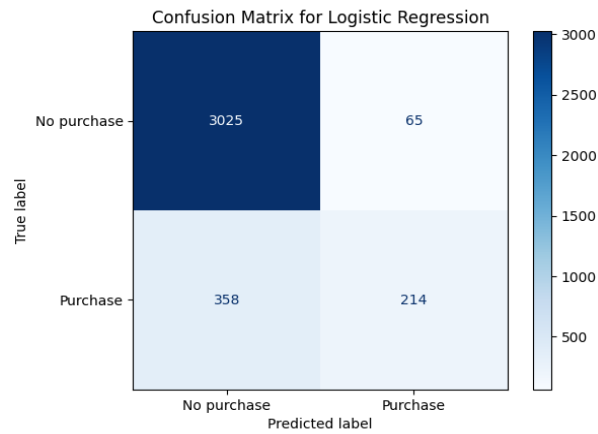
SVM



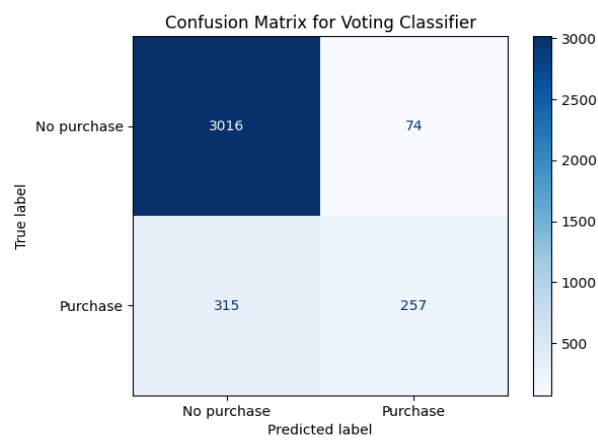
XGBoost



Random Forest

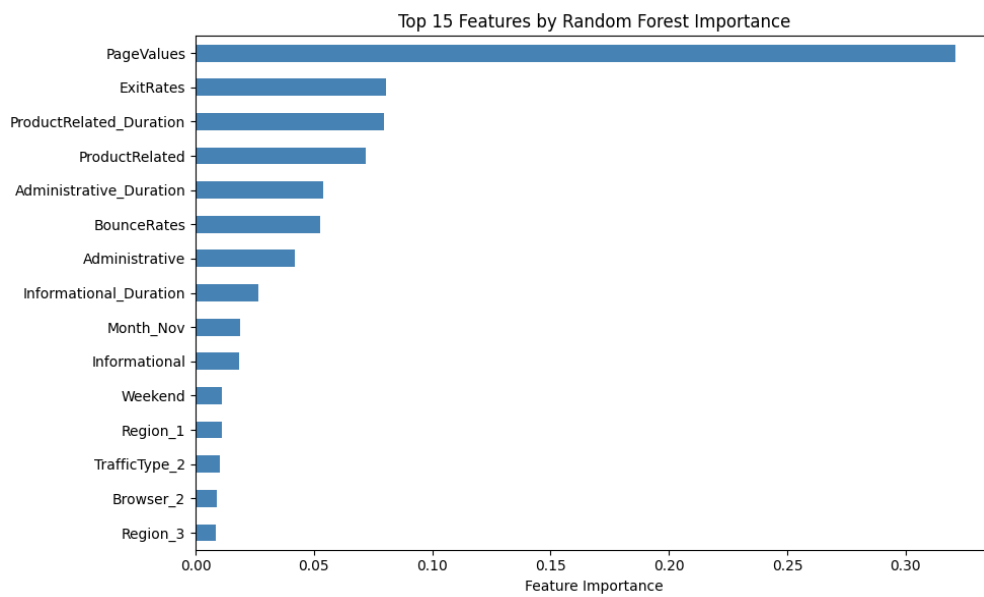


Logistic Regression



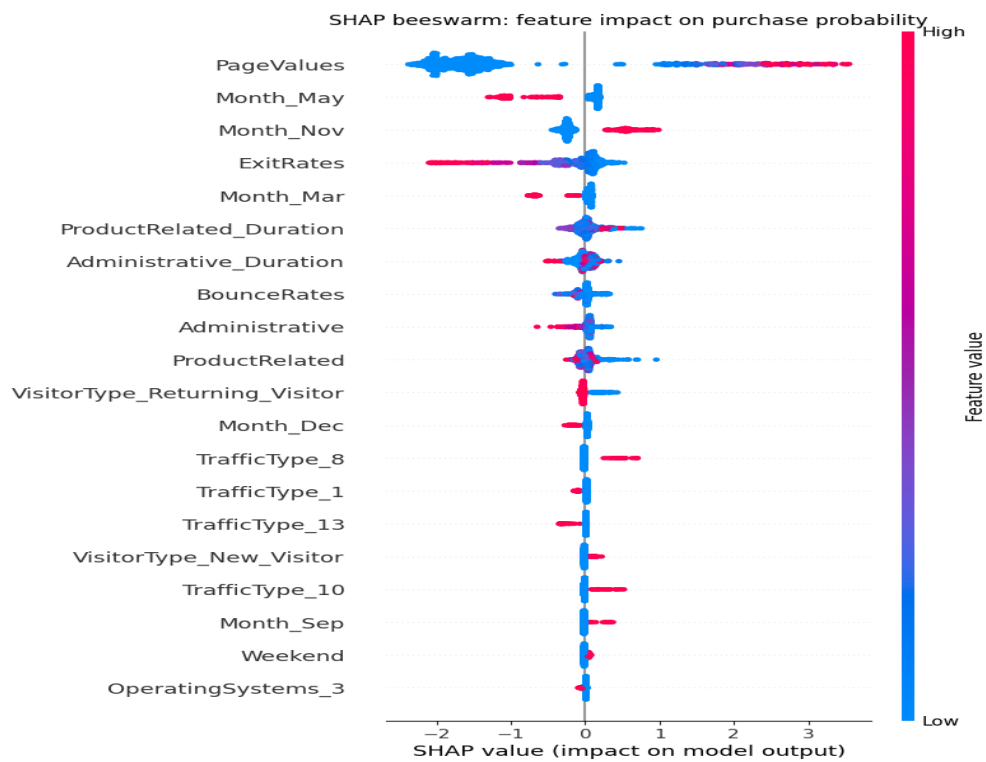
Voting Classifier

Appendix 2 - Random Forest Feature Importance



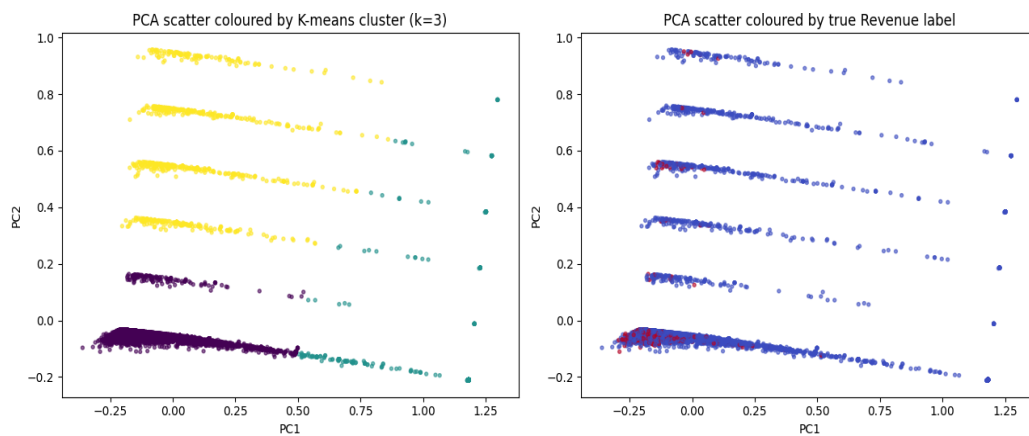
Top 15 features by Random Forest importance.

Appendix 3 - SHAP Beeswarm Summary



SHAP beeswarm summary for the tuned XGBoost classifier.

Appendix 4 - PCA Projection with K-means Clusters



PCA projection coloured by K-means cluster and by true Revenue label.